Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Perchonok, Michele Ph.D.		
Project Title:	Bulk Ingredient-Based Menu Development		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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PI Web Page:	http://hefd.jsc.nasa.gov/		
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The National Aeronautics and Space Administration (NASA) is working towards future long duration manned space flights beyond low earth orbit. The duration of these missions may be as long as 2.5 years and will likely include a stay on a lunar or planetary surface. The primary goal of the Advanced Food System in these long duration exploratory missions is to provide the crew with a palatable, nutritious, and safe food system while minimizing volume, mass, and waste. During the stay on the lunar or planetary surface, food processing could occur to provide the crew with more variety and freshly produced foods. The question has been posed on which crops should be locally grown and which crops should be shipped as staple, long shelf-life ingredients. Some specialized packaging and/or storage conditions may be required to meet the 3 to 5 year shelf life. These staple items would likely be shipped in bulk packaging, allowing for some savings in storage volume.		

Task Description:	The objective of this project is to develop a bulk ingredient-based menu. The menu will consider previous vegetarian recipes and menus that have been developed for Advanced Life Support (ALS). The menu will be comprised of staple ingredients that can be stored for up to 5 years. These bulk ingredients will be further processed into edible ingredients and then used to prepare menu items. The menu will be supplemented with vegetable crops from the ALS approved list. Based on the menu, the quantities of bulk ingredients and fresh vegetables will be calculated. However, several calculations or estimates need to be made, prior to obtaining the final bulk quantities. First, the mass of the inedible portion of the staple ingredient, such as the hull, will be estimated. Next, the processing yields will be determined for each of the ingredients produced. Finally, any loss experienced during the food preparation step, such as moisture, will be calculated. In all calculations, a 600-day mission with a crew of six will be used.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A collection of vegetarian based recipes will be available within a very understandable database.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 01/30/2012)